
Evidence-to-Law Credibility Assessment

Framework: ASME V&V 40-2018

Question of interest: is the gate credible for clearing robot code?

Context of use: pre-execution clearance in a Physical AI oncology trial

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Abstract.

This assessment template applies a risk-informed credibility framework, in which model risk is influence times consequence, to the verify-before-generate gate.

Disclaimer: This work is independent and not endorsed or sponsored by trial sponsors, FDA, CROs, sites, IRBs, regulators, or medical societies; and was generated using Artificial Intelligence.

Keywords

1 Question of interest and context of use

The question is whether the automated gate is credible enough to clear robot-patient interaction code before execution. The context of use is high-consequence and high-influence, so the credibility bar is set accordingly.

2 Risk: influence times consequence

Because the gate’s output directly governs a physical motion, model influence is maximal and decision consequence is severe, placing the assessment in the highest-rigor regime of the framework.

Figure placeholder (Images/risk-matrix.png). The float, caption, and \label mirror the VVUQ-02 figures; replace the box with \includegraphics.

Figure 1: Figure 1. Placeholder influence-by-consequence risk matrix.

3 Credibility goals

Goals follow from the risk placement.

verification exhaustive; fraction equals 1.0

validation agreement against named standards with a floor per gate
uncertainty run-to-run variation bounded by a coefficient-of-variation limit

4 Credibility evidence

Factor	Goal	Evidence
verification	fraction 1.0	51/51; 172/172
validation	agreement floor	per-gate agreement
uncertainty	CoV bound	32/32 sweep clear

Table 1: Table 1. Credibility factors, goals, and evidence (modeled on VVUQ-05 Table 1).

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Ethical Disclosures

The author declares no competing interests. This is an independent draft and is not endorsed, sponsored, or approved by any trial sponsor, CRO, site, IRB, regulator, or medical society, nor by CFR, ICH, or FDA. All supporting numbers are simulation or illustrative results.

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Data Availability

Supporting records are openly available in the [kevinkawchak/cancer-automated](#) and [kevinkawchak/physical-ai-oncology-trials](#) GitHub repositories and are archived on Zenodo. All data sets are synthetic or illustrative and reproducible from a deterministic seed; no patient-identifiable information is included.

References

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